

MATH 2790 Fall 2025

Homework 11

Due date: Wednesday, Nov 19

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 6.2–6.4.

Problems:

- 1. Question (1) on page 317 of textbook under section 6.2.
- 2. Question (2) on page 317 of textbook under section 6.2.
- 3. Question (3) on page 317 of textbook under section 6.2.
- 4. Question (1) on page 318 of textbook under section 6.3.
- 5. Question (2) on page 318 of textbook under section 6.3.
- 6. Question (1) on page 319 of textbook under section 6.4.
- 7. Question (2) on page 319 of textbook under section 6.4.
- 8. A \$1000 face value 20-year bond with annual coupons and redeemable at maturity at \$1050 is purchased for P to yield an annual effective rate of 8.25%. The first coupon is \$75. Each subsequent coupon is 3% greater than the preceding coupon. Determine P . (Hint: fully understand how the payments are made and review the required knowledge of non-level annuity to find the price.)